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Maternally inherited hearing loss, ataxia and myoclonus associated with a novel point mutation in mitochondrial tRNA$^{Ser(UCN)}$ gene

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We report on a new maternally-inherited syndrome characterized by a combination of sensorineural hearing loss, ataxia and myoclonus in a large kindred from Sicily. Hearing loss was the most widespread and sometimes the only symptom found in family members. Sequence analysis of the mitochondrial DNA regions encompassing the tRNA genes revealed the presence of a heteroplasmic insertion at nucleotide position 7472. The insertion adds a seventh cytosine to a six-cytosine run that is part of the mitochondrial tRNA$^{Ser(UCN)}$ gene. Conformational analysis showed that this mutation is likely to alter the structure of the TΨC loop in the tRNA$^{Ser(UCN)}$ clover leaf secondary structure. Moreover, the degree of heteroplasmy in blood and muscle was correlated with the clinical phenotype, and homoplasmic mutant cybrids showed decreased complex I activity, low oxygen consumption and high lactic acid output, indicating faulty oxidative phosphorylation. Finally, mutation was absent in 381 unrelated maternal lineages, suggesting specific segregation with the disease. We propose that the C7472 insertion-mutation is pathogenic, and etiologically related to hearing loss and other symptoms that define a novel maternally-inherited clinical entity.

INTRODUCTION

An increasing number of disorders associated with mutations of the mitochondrial genome have been reported in recent years (1,2). In general, specific mutations are consistently related to well-defined clinical entities, suggesting the presence of a pathogenetic link between molecular and clinical abnormalities. For instance, large-scale rearrangements of mitochondrial DNA (mtDNA) are almost invariably associated with an ocular myopathy, Chronic Progressive External Ophthalmoplegia, with or without the additional symptoms of Kearns-Sayre syndrome. On the other hand, a collection of specific mtDNA point mutations are responsible for most of the maternally-inherited syndromes, including Leber's Hereditary Optic Neuropathy (LHON); Neurogenic Ataxia and Retinitis Pigmentosa (NARP); non-syndromic acute aminoglycoside-induced deafness (AAID); Myoclonic Epilepsy with Ragged Red Fibres (MERRF); Maternally Inherited Myopathy and Cardiomyopathy (MMC); Mitochondrial Encephalomyopathy with Lactic Acidosis and Stroke-like Episodes (MELAS).

Sensorineural deafness and cerebellar signs are frequently observed in mitochondrial disorders (3,4). Here, we describe a mitochondrial syndrome characterized by progressive hearing loss, either isolated or in combination with ataxia, dysarthria, and, more rarely, focal myoclonus, that was found in several maternal members of a large Italian pedigree. Sequence analysis of mitochondrial DNA in a proband revealed the presence of a novel point mutation consisting in the insertion of an extra-cytosine (C) nucleotide in a six-C stretch that is part of the gene encoding tRNA$^{Ser(UCN)}$. We provide evidence that this mutation is etiologically bound to the phenotype, because of the following arguments: (i) the mutation was heteroplasmic; (ii) the degree of heteroplasmy correlated with the disease phenotype; (iii) the mutation is likely to introduce a functionally relevant conformational change in the tRNA cloverleaf structure; (iv) cybrids obtained by repopulating mtDNA-less cells with the mutant mitochondrial genome were respiration-deficient, in contrast with cybrids containing the wild-type mitochondrial genome derived from the same individual. The identification of a new mtDNA mutation associated with syndromic and non-syndromic hearing loss corroborates the view that analysis of the mitochondrial genome should be included in the diagnostic approach to genetic deafness, a frequent and highly disabling condition.

CLINICAL STUDIES

The family tree is shown in Figure 1. The family originated from Sicily; the maternal grandparents of the proband were first cousins. No record of these subjects is presently available.

The proband (III-12), a 34 year old male, has been suffering progressive bilateral hearing loss with tinnitus since age 25. Since age 28 he presented focal myoclonic jerks of the right abdominal muscles with no corresponding EEG abnormalities. Clinical examination revealed the presence of an ataxic gait,

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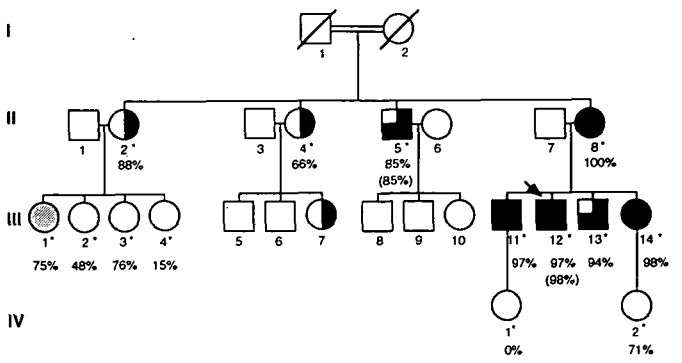

Figure 1. Family tree. Black solid symbols indicate severely affected individuals. Half-black symbols indicate individuals affected by hearing loss only. Three-quarter black symbols indicate individuals affected by a syndrome of intermediate severity. Open symbols indicate unaffected, reportedly unaffected, or clinically unknown individuals. The shaded symbol refers to an individual with neurological symptoms attributed to multiple sclerosis (see text). Asterisks indicate individuals that were clinically examined and whose DNA was analyzed. Numbers in percentage indicate the relative proportion of mutant mtDNA in lymphocytes; the bracketed percentages indicate the relative proportion of mutant mtDNA in muscle. An arrow indicates the proband.

hypodiadochokinesia, dysarthria, and mild generalized hypotonia with absence of deep tendon reflexes. Bilateral dysesthesia with a distal ('sock') distribution was observed in the lower limbs. Somatosensory evoked potentials (SEPs) showed mild abnormalities in both the peripheral and central components of the somatosensory pathways. Audiometric examination showed an average hearing loss of 45 decibels (dB), more severe for high frequencies. An audiometric recruitment test was positive, indicating compromission of the cochlear receptor. Brainstem auditory evoked potentials (BAEPs) showed cochlear and retro-cochlear abnormalities. Blood lactate, pyruvate and alanine at rest were abnormally elevated [lactate: 3204 μmol/l, normal values (n.v.) 500–1800; pyruvate: 202 μmol/l, n.v. 60–130; alanine: 620 μmol/l, n.v. 150–360]. Moreover, blood lactate and pyruvate increased markedly during and after a standard physical exercise, indicating an important block of the aerobic utilization of pyruvate. The optic fundi and ECG were normal. CT and MRI examinations showed a moderately enlarged fourth ventricle and mesencephalic aqueduct, and atrophy of superior part of the cerebellar vermis, while the other regions of the brain were normal.

The 60 year old mother of the proband (II-8) had deafness and tinnitus from age 15. On clinical examination, she was dysarthric, ataxic and hypotonic. Audiometric examination revealed a lesion similar to that found in the proband. Bilateral carpal tunnel syndrome and mild abnormalities of the somatosensory pathway were observed by electromyography. A 37 year old brother (III-11) complained of moderate hearing loss from age 28; tonal audiometry and BAEPs indicated a bilateral retrocochlear lesion; he had ataxia, adiadochokinesia and dysarthria; he was hypotonic and deep tendon reflexes were absent. A 'sock' dysesthesia was present bilaterally, and he reported the occurrence of occasional myoclonic jerks of the abdominal muscles. Blood lactate and pyruvate were markedly increased during and after exercise. Another brother, (III-13), age 32, had mild neurosensorial hearing loss, documented by an abnormal audiometric curve. A 30 year old sister (III-14) presented severe hearing loss and dysarthria since she was 13, and tremor since she was 25. She was ataxic, hypotonic and complained of dysesthesia with a 'sock' distribution in the lower extremities. BAEPs and audiometry showed a mixed cochlear and retro-cochlear lesion.

A 65 year old uncle of the proband (II-5) presented moderate hearing loss, a mild ataxic gait, and bilateral 'sock' dysesthesia. Blood lactate, pyruvate and alanine were within the normal range. Electromyography was normal but SEPs revealed a mild polyneuropathy. His two sons and his daughter (III-8, III-9, III-10) were referred to as normal.

The 62 and 67 year old aunts (II-2 and II-3) reported the presence of hearing loss, tinnitus, and episodes of 'vertigo'. Audiometric examination was refused.

Individual III-7, the 39 year old daughter of individual II-3, was referred to as having isolated, mild hearing loss. No information was available on her two brothers (III-5 and III-6).

Of the four daughters of individual II-2, one (III-1) has been suffering from age 25 of a neurological syndrome characterized by optic neuritis, ataxia, diplopia and moderate mental deterioration. On clinical examination, ataxia, spontaneous nystagmus, and hyperreflexia were noticed. No hearing loss, tinnitus, or vertigo were referred to by the patient. A diagnosis of multiple sclerosis was given on the basis of relapsing course, high levels of oligoclonal immunoglobulins in the CSF, responsiveness to corticosteroids, and identification of numerous demyelinating areas on MRI examination of the brain. Her three sisters (III-2, III-3, III-4) reported no symptoms, and were normal on clinical examination. Two nieces of the proband, (IV-1 and IV-2), both 4 years old, were normal as well.

RESULTS

Morphological and biochemical studies

Light microscopy examination of a muscle biopsy from the proband (III-12) failed to show the presence of ragged-red fibers. However, there was lipid accumulation, and the histochemical reaction to cytochrome *c* oxidase (COX) was barely detectable in 48% of the fibers. Electron-microscopy examination confirmed the presence of altered mitochondria with concentric cristae and numerous paracrystalline inclusions. The muscle respiratory chain enzyme activities showed marked reduction in complex I and COX (complex IV) activities, normalized to citrate synthase (CS) (complex I/CS = 4.4; normal range: 9.4–24.2; complex IV/CS = 14; normal range: 39–141). Other results were within normal range.

A muscle biopsy obtained from individual II-5 showed moderate neuropathic changes with no ragged-red fibers. Histochemical COX deficiency was present in 3% of the fibers. Activities of the respiratory chain enzymes normalized to CS were normal in muscle homogenate.

Mitochondrial DNA analysis

The clinical features, the morphological and biochemical results from the proband's muscle biopsy, and the transmission of the syndrome suggesting maternal inheritance, prompted us to analyze the proband's mitochondrial genome. Known

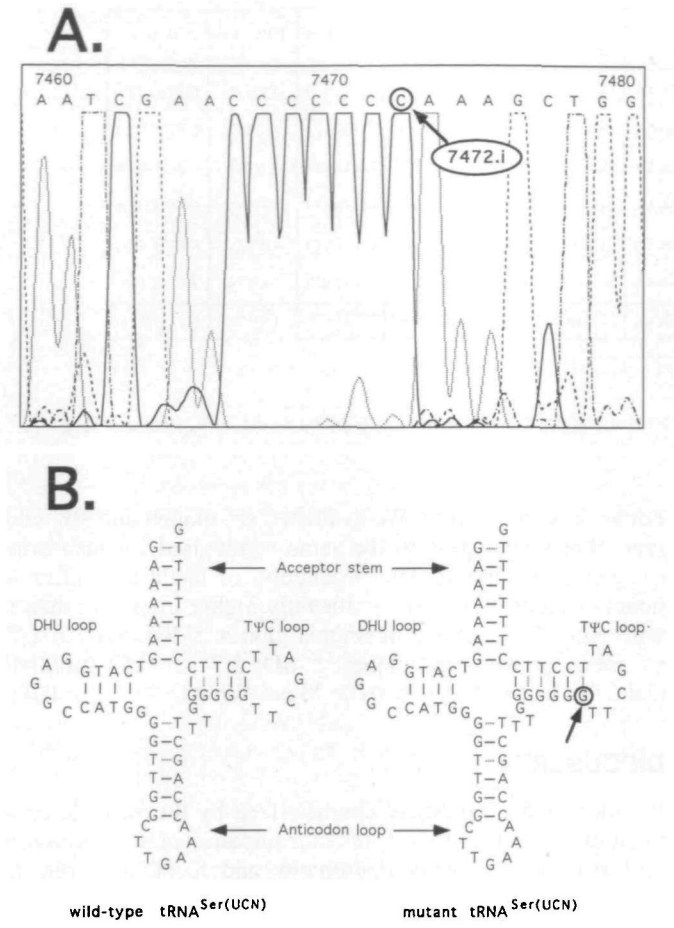

Figure 2. Point mutation in mitochondrial $tRNA^{Ser(UCN)}$. (**A**) Automated sequence analysis of the region encompassing the $C^{7472.i}$ mutation (arrow) of the proband's mtDNA. Peaks corresponding to each nucleotide are indicated by different line patterns. (**B**) Proposed secondary structures of the wild-type and mutant $tRNA^{Ser(UCN)}$ gene transcripts. In the mutant, the TΨC stem is extended and the TΨC loop is shortened, as compared to the wild-type.

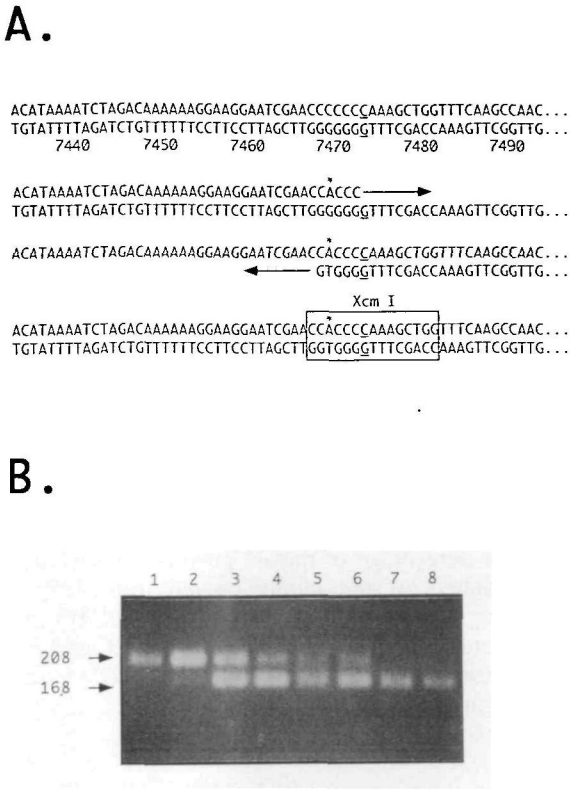

Figure 3. Creation of a *Xcm*I site diagnostic for the $C^{7472.i}$ insertion of mtDNA $tRNA^{Ser(UCN)}$ gene. (**A**) Scheme of the PCR amplification performed on the mtDNA template containing the additional $C^{7472.i}$ (underlined). An asterisk indicates the change to the corresponding Cambridge sequence introduced in the sense primer (see Methods). (**B**) Diagnostic digestion with *Xcm*I. In the presence of the $C^{7472.i}$ mutation, a 208 bp PCR DNA fragment (arrow) is cut into a 168 bp fragment (arrow) and a 40 bp fragment (not shown). *Lane 1*: subject IV-1; *lane 2*: subject III-4; *lane 3*: subject III-2; *lane 4*: subject III-1; *lane 5*: subject II-5; *lane 6*: subject IV-2; *lane 7*: subject III-12; *lane 8*: subject II-8.

mtDNA mutations and large-scale rearrangements of mtDNA were ruled out by suitable RFLP- and Southern-blot analyses. In particular, we failed to detect the following mutations: the G^{1555} and C^{7445} mutations associated with AAID and non-syndromic spontaneous deafness, respectively; the G^{3243} and G^{3271} 'MELAS' mutations; the G^{8344} and C^{8356} 'MERRF' mutations; the G^{3260} 'MMC' mutation; the G^{8993} and C^{8993} 'NARP' mutations; the A^{11778}, A^{3460}, A^{15257}, C^{14484} 'LHON' mutations (1,2). Since most of the pathogenic mtDNA mutations are concentrated in the tRNA genes, sequence analysis was begun on PCR-amplified proband's mtDNA segments encompassing these genes. Direct sequence analysis failed to detect any change from the 'Cambridge' mtDNA sequence (5), with the exception of an extra C that extends by one nucleotide a six-C stretch (mtDNA replication sense) encompassing nucleotide positions (np) 7466–7471 (Fig. 2A). We termed the mutation as $C^{7472.i}$, where 'i' stands for 'insertion'. However, it is important to remember that since the $tRNA^{Ser(UCN)}$ gene is located in the strand complementary to that published in the 'Cambridge' sequence, the C-stretch in the latter corresponds to a G-stretch in the TΨC stem of the $tRNA^{Ser(UCN)}$ transcript (Fig. 2B).

The additional G can be placed in either the TΨC stem or in the variable arm of the $tRNA^{Ser(UCN)}$ clover leaf. However, according to the Zucker algorhythm for RNA folding, the first configuration is thermodynamically more favourable ($\Delta G_{(T\Psi Cstem)} = -3.2$ kCal/mol vs. $\Delta G_{(variable\ arm)} = -2.1$ kCal/mol), as it induces the stabilization of the secondary structure by creating a pairing between the extra G with the T at np 7478. This conformation creates a distortion of the TΨC-loop, that results in six bases instead of seven, while the TΨC stem is extended by one pair (six instead of five) (Fig. 2B).

The possibility of multiple length variations of the region was evaluated by sequence analysis of 10 recombinant plasmids containing a 218 bp PCR insert corresponding to the mtDNA sequence from nt 7362 to nt 7580 derived from the proband's DNA, and one plasmid containing the corresponding fragment from a control DNA (see Methods). The 11 clones resulted identical to the corresponding Cambridge sequence, except that all the clones from the proband contained the $C^{7472.i}$

	Acceptor stem		DHU stem	DHU loop	DHU stem		Anticodon stem	Anticodon loop	Anticodon stem	Variable arm	TΨC stem	TΨC loop	TΨC stem	Acceptor stem
Human	GAAAAAG	T	CATG	GAGGC	CATG	G	GGTTGG	CTTGAAA	CCAGCT	TTG	GGGGG	TTCGATT	CCTTC	CTTTTTTG
Rat	GAGAAAG	A	CATA	ATGGT	TATG	A	AGTTGG	CTTGAAA	CCAGTT	GTA	GGGGG	TTCGATT	CCTTC	CTTTCTTA
Mouse	GAGAAAG	A	CATA	TAGGA	TATG	A	GATTGG	CTTGAAA	CCAATT	TTA	GGGGG	TTCGATT	CCTTC	CTTTCTTA
Bovine	GAGAGAG	A	CATA	GAGGT	TATG	A	TGTTGG	CTTGAAA	CCAATA	GTA	GGGGG	TTCGATT	CCTTC	CTTTCTTA
Chicken	AAGAAAG	AA	GCAT	TAAGTGGTTTG	ATGC	G	GTTGG	CTTGAAA	CCAAC	ATGT	GAGGG	TTCGATT	CCTTC	CTTTCTTG
X. laevis	AAGAAAG	TG	GCA	GAGTGGTGA	TGC	A	ACTGA	CTTGAAA	TCAGA	GTAG	GGGGG	TTCGATT	CCCTC	TTTTCTCG
Mutant	GAAAAAG	T	CATG	GAGGC	CATG	G	GGTTGG	CTTGAAA	CCAGCT	TTG	GGGGGG	TTCGATT	CCTTC	CTTTTTTG

Figure 4. Alignment of $tRNA^{Ser(UCN)}$ genes from different species. Notice the high conservation of sequences corresponding to the TΨC stem and loop.

mutation, making the hypothesis of multiple length variations very unlikely .

Finally, the percentage of mutant genomes observed in the maternal members of the family ranged from 15% in an asymptomatic individual (III-4) to virtually 100% in the severely affected mother (II-8) of the proband (Figs 1,3). The $C^{7472.i}$ was not detected in any of the 381 DNA samples analyzed as controls.

Cell culture

Generation of transmitochondrial cybrids. We collected 62 transformant cybrid clones after fusion of rho° cells derived from a clonal 143B cell line with cytoplasts obtained by enucleation of myoblasts of Patient III-12. Densitometry on *Xcm*I-digested mispaired-PCR products, and solid-phase mini-sequencing (SPMS) were used to evaluate the percentage of wild-type and mutant genomes. A total of 58 clones were homoplasmic for the mutant mtDNA species; only two clones were homoplasmic for the wild-type mtDNA. To evaluate the variability of the wild-type phenotype, several subclones were obtained from the two wild-type clones. Two additional clones were initially heteroplasmic, as they had 60% and 80% mutant mtDNA species in the first measurement, but in a second assay performed two weeks later they had become homoplasmic mutant.

Biochemical studies. Specific activities were measured in 13 homoplasmic mutant cybrids and seven wild-type clones. A significant decrease of Complex I activity was observed in mutant clones compared to wild-type clones (mean of mutant clones ± SD: 9.4 ± 2.0 μmol/min/g vs. mean of wild-type clones ± SD: 13.1 ± 2.1; Student's t test $p = 0.001$; Mann-Whitney test $p = 0.005$). By contrast, COX activities were not significantly different between the two groups (mean of mutant clones 20.6 ± 9.4 μmol/min/g vs. mean of wild-type clones 26.9 ± 13.0).

Oxygen consumption. We evaluated 11 clones harboring the mutation and compared them with seven wild-type clones. Oxygen consumption was lower in the clones harboring the mutation than in wild-type clones (mean of mutant clones ± SD: 1.91 ± 0.21 femtomole/min/cell vs. mean of wild-type clones ± SD: 2.61 ± 0.72; Student's t test $p = 0.01$; Mann-Whitney test $p = 0.02$).

Lactic acid production. We evaluated six mutant and six wild-type clones belonging to the same series used for measuring oxygen consumption. The production of lactic acid after 48 hours of incubation was significantly higher in mutant than in wild-type clones (mean of mutant clones ± SD: 4.47 ± 1.77 vs. mean of wild-type clones ± SD: 2.74 ± 0.67 μmol/10^6 cells; Student's t test $p < 0.05$; Mann-Whitney test $p = 0.02$).

DISCUSSION

We identified a syndrome characterized by the variable combination of hearing loss, cerebellar impairment and, occasionally, peripheral sensory neuropathy and focal, non-epileptic myoclonus. The clinical phenotype was present only in the maternal lineage of a large two-generation pedigree, suggesting an mtDNA origin of the responsible mutation. A novel mtDNA change was indeed found during sequence analysis of the mitochondrial tRNA genes of a proband. The mutation consists in the insertion of an extra C ($C^{7472.i}$) in a six-C stretch. The six C-run in the published 'Cambridge' sequence corresponds to a poly-guanosine stretch in the $tRNA^{Ser(UCN)}$ gene and transcript. Interestingly, cytosine-rich sequences are thought to predispose to length variations in mtDNA of humans and other primates (6). Polypyrimidine stretches have the potential to form bent DNA (7), that is prone to spontaneously expose its single strands and thus allow slipped mispairing to occur. Likewise, mtDNA deletions or insertions in human mitochondrial myopathies usually occur across polypyrimidine-rich direct repeats scattered throughout the mtDNA molecule (8,9) and a 7 bp $A(C)_6$ sequence forms the bridging element of a heteroplasmic tandem duplication in the D-loop, the pathogenic significance of which remains unclear (10,11). However, since the mitochondrial genome has a mutation rate that is remarkably higher than that of the nuclear genome, the identification of a rare polymorphism in an affected individual is not sufficient to establish a causative link. The following additional features of the $C^{7472.i}$ change substantiate its pathogenic role. First, the mutation was specifically associated with the affected maternal lineage, as it was absent in 381 unrelated controls. Second, the mutation introduces a conformational change that is likely to affect the supposed functional role of the TΨC loop in specifying the tRNA contact to the small ribosomal subunit during translation. In keeping with this, we notice that the base composition and structure of the $tRNA^{Ser(UCN)}$ TΨC loop

is highly conserved among species (12) (Fig. 4), suggesting that their perturbation may be deleterious. Third, the mutation was heteroplasmic. Albeit not obligatory, heteroplasmy is considered as highly indicative for pathogenicity, as it suggests that the mutation occurred recently, and that a mechanism of negative selection is likely to act against the stable fixation of the mutation in a given mitochondrial haplogroup (13). In addition, the presence of heteroplasmy offers a unique tool to investigate the effects of the mutation quantitatively, because it makes it possible to correlate the percentage of mutation with the severity of the clinical presentation, and with biochemical or physiometabolic indexes of faulty oxidative phosphorylation (OXPHOS). In our case, combined deficiency of complex I and IV was present in the proband's muscle, that contained 97% mtDNA of the mutant species; by contrast, no biochemical defect was measurable in the muscle of individual II-5, containing 85% mutant mtDNA. Interestingly, in these two subjects the relative amount of mutant mtDNA in muscle was comparable with that found in blood, suggesting that quantification of heteroplasmy in blood may adequately reflect the degree of heteroplasmy in other tissues, including muscle and, possibly, brain. Keeping these considerations in mind, we observe that in the most severely affected individuals, i.e. subject II-8 and her offsprings, III-11, III-12, III-13, and III-14, the percentage of mutant mtDNA ranged from 94% (individual III-13) to virtually 100% (individual II-8) (Figs 1,3B). By contrast, the moderately affected individual II-5 had 85% mutant mtDNA in blood; his sisters, II-2 and II-3, who complained of isolated, non-syndromic hearing loss with tinnitus, had 88% and 66% mutant mtDNA, respectively. Likewise, the relative amount of mutation in the unaffected maternal members of the family that could be investigated ranged from 15% to 76%. A special mention should be made of subjects III-1 and IV-2. The first subject (III-1) was diagnosed as suffering from multiple sclerosis on the basis of multiple and concordant clinical and laboratory features. A 75% mutant mtDNA was demonstrated in her blood. Whether the presence of the mutation can have triggered or worsened the course of multiple sclerosis in this case remains an interesting but as yet unproven hypothesis. The second subject (IV-2) showed the presence of 71% mutant mtDNA in blood; she is presently asymptomatic, but her young age (4 years) makes any prognostic evaluation difficult.

These data suggest the presence of a positive, although loose, correlation between the relative amount of mutation and the clinical presentation. However, the pathogenic potential of the mutation must be relatively low, since the threshold above which the full-blown syndrome is manifested is very high (≈95%). Moreover, in contrast with what is observed for highly pathogenic mutations (14), the degree of heteroplasmy detected in a rapid-turnover tissue such as blood was not substantially lower than that found in a stable, post-mitotic, highly differentiated tissue such as skeletal muscle.

To provide further evidence of the deleterious effects of the mutation on OXPHOS metabolism, we prepared trans-mitochondrial cybrids by fusing 143B-derived rho° cells with enucleated myoblasts obtained from the proband's muscle. The use of transmitochondrial cybrids allows one to study a given mtDNA mutation without the influence of the autochthonous nuclear background (15), making it possible to determine whether the mutation can cause *per se* a functional or biochemical defect in the cell (16–18). The reliability of this system is further improved when, as in this case, the mutation is heteroplasmic, because the effects of the mutant mtDNA species can then be compared with those produced by a haploidentical wild-type species derived from the same individual.

Characterization of the $C^{7472.i}$ by RFLP analysis of mispaired PCR products and SPMS showed that the vast majority (97%) of our cybrid clones were homoplasmic mutant. This is not surprising if we consider that the amount of mutation in the donor's muscle mtDNA was identical (97%) to the percentage of homoplasmic mutant cybrid clones. If we assign an average of five genomes per organelle, and hypothesize an even distribution of mutant and wild-type genomes in muscle, only one out of six mitochondria will contain one wild-type genome, whereas the remaining five will exclusively harbor mutant genomes. However, an additional mechanism favoring the preferential amplification of the mutant species is suggested by the observation that the only two clones of our collection that were heteroplasmic underwent a rapid drift towards mutant homoplasmy. This phenomenon has been already reported, and more systematically studied, in clones harboring the G^{3243} 'MELAS' mutation (19).

In order to make it possible to compare our several homoplasmic mutant clones with an adequate number of wild-type clones, we generated five subclones from the only two homoplasmic wild-type clones obtained after cybrid fusion. We reasoned that since all cybrids derived from a single 143B rho° clone (18), and all subclones were homoplasmic wild-type, they were genetically indistinguishable from the parental clones, and could then be used to assess the range of biochemical and metabolic variability associated with the 'normal' mitochondrial genome of the proband.

In our cybrid-cell system, the presence and the amount of the $C^{7472.i}$ mutation did correlate with defective mitochondrial respiration, examined by analysis of complex I activity, oxygen consumption rate and lactic acid production. Complex I activity is exquisitely sensitive to defects of mtDNA, since seven of the 13 mitochondrial protein-encoding genes specify as many subunits of this respiratory enzyme. Oxygen consumption in intact cells is a good index of the coordinated activity of the respiratory chain complexes. Increase of lactic acid levels is usually due to a block in the aerobic utilization of pyruvate. However, the absence of significant abnormalities of individual complex IV activity in mutant cybrids suggests that the $C^{7472.i}$ mutation is moderately pathogenic. Since both oxygen consumption and lactic acid production depend upon the overall aerobic capacity of the cell, one possibility is that multiple moderate defects of the respiratory enzymes can synergistically combine, resulting in more marked alteration of these 'global' indicators of aerobic metabolism.

Based on the above considerations and on our experimental results, we conclude that the $C^{7472.i}$ mutation is indeed pathogenic, and that its presence alone is sufficient to explain the OXPHOS defects found in our patients *ex vivo* and, presumably, *in vivo* as well.

From a clinical standpoint, three aspects of this syndrome are particularly noteworthy. First, symptoms were predominantly affecting the central and peripheral nervous systems, with preferential involvement of structures in the posterior fossa (the cerebellum and the cochlear nuclei of the brainstem),

while other organs, including the skeletal muscle and the heart, were spared. In addition, no ragged-red fibres were present in muscle, in spite of the presence of lipid accumulation and partial COX deficiency. This observation adds further support to the notion that mitochondrial disorders are extremely variable in clinical presentation, and prompts one to consider the possibility of a mitochondrial etiology also in unexplained clinical syndromes that lack some or most of the morphological hallmarks of a mitochondrial encephalomyopathy.

Second, some unusual features were found identical in different maternal members of the family. For instance, individuals III-12 and III-13 were affected by a rare form of focal myoclonus of subcortical, possibly spinal, origin, that involved the same muscle group in both subjects. Likewise, several patients suffered from a 'pure' sensory neuropathy of the lower limbs. Whether these features are due to the mutation itself, or to the mitochondrial or nuclear genetic backgrounds of the family is unclear. The identification of other maternal lineages harboring the same $C^{7472.i}$ mutation will help understand its specific contribution to the clinical phenotype.

Third, we notice that the earliest and most widespread symptom in the family was hearing loss. In several members hearing loss was indeed the only symptom, so that an early diagnosis of non-syndromic neurosensorial deafness was given to these subjects, before a complete study of the family could lead to the identification of additional symptoms occurring in the most severely affected members of the kindred. Hearing loss is a common, albeit overlooked feature of mitochondrial disorders. In most cases it is part of complex syndromes in which neurological, muscular or cardiac abnormalities produce the most prominent and life-threatening symptoms. In other cases, hearing loss may be present alone, or in association with non-neurological abnormalities such as diabetes mellitus. Maternally inherited acute aminoglycoside-induced non-syndromic deafness has been described in several families worldwide, in association with a mtDNA mutation at nt 1555 in the 12S rRNA gene (20). A second mtDNA mutation has been associated with spontaneous non-syndromic maternally-inherited deafness in a large Scottish family (21). Interestingly, this mutation affects nt 7445, that is considered as either the terminal position at the 3′ end of $tRNA^{Ser(UCN)}$ transcript (5), or the position immediately next to the corresponding gene (12). Therefore, the same tRNA gene can be affected by two different mutations, the C^{7445} and the $C^{7472.i}$, that are exclusively or predominantly associated with neurosensorial hearing loss. Whether this is a merely circumstantial observation or a pathogenetically relevant phenomenon is presently unknown.

In any case, this study supports the idea that analysis of mtDNA should be included in the diagnostic approach to inherited neurosensorial hearing loss, a condition of great social relevance and emotional impact, especially when the family study shows compatibility with maternal transmission of the trait.

MATERIALS AND METHODS

Muscle biopsies, morphological and biochemical studies

Muscle needle biopsies of the left quadriceps were obtained after informed consent from the proband, individual III-12, and his uncle, individual II-5. For morphological examination, muscle tissue was rapidly frozen in liquid-nitrogen cooled isopentane and maintained in liquid nitrogen until use. Standard techniques (22) were used for histological, histochemical and electron microscopy studies.

Rotenone-sensitive NADH-Coenzyme Q1 reductase (complex I), succinate dehydrogenase (part of complex II), cytochrome *c* oxidase (COX, complex IV) (23), antimycin-sensitive ubiquinol cytochrome *c* reductase (complex III) (24), and citrate synthase (CS) (25) were evaluated in muscle homogenates, as well as in frozen-thawed, mildly-sonicated, cybrid cells.

Oxygen consumption on intact cells was measured with an oxygraph (Hansatech, King's Lynn, UK), as described (18,25).

Lactic acid was measured (26) 48 h after plating 5×10^5 cells in 60 mm Petri-dishes, as described (18).

DNA studies

Total DNA was extracted from muscle (III-12 and II-5) and blood lymphocytes (II-2, II-4, II-5, II-8, III-1-, III-2, III-3, III-4, III-5, III-11, III-12, III-13, III-14, IV-3, IV-4) as previously described (27). The mtDNA regions encompassing the 22 tRNA genes were PCR-amplified from the proband's muscle DNA using suitable pairs of primers. For each PCR fragment, direct sequencing was performed on both strands using a 373A automatic DNA sequencer (Applied Biosystems). Suitable RFLP- and Southern-blot analyses were used to rule out the presence of known mutations associated with mitochondrial disorders.

Mispaired PCR (28) and solid-phase minisequencing (29) (SPMS), were used to verify the presence, and quantify the amount of mutation found in the proband's mtDNA in all of the available maternal family members and in controls.

In mispaired PCR (Fig. 3A), the detection of the C insertion was obtained by modifying the sense primer, corresponding to mtDNA positions 7432–7471, by replacing the wild-type G with an A (underlined). This change creates a diagnostic *Xcm*I restriction site ($CCA(N)_9TGG$) at position 7472 when the mutant sequence is amplified (5′-ACATAAAATCTAGA-CAAAAAAGGAAGGAATCGAACCACCC-3′). The modified sense primer was used in combination with an antisense primer corresponding to positions 7640–7616. In the presence of the mutant extra-C, *Xcm*I cuts the 208 bp PCR product into two 168 bp and 40 bp fragments, which can be easily differentiated from the undigested wild-type fragment by electrophoresis through a 3% agarose gel in TAE buffer followed by ethidium-bromide staining (Fig. 3B).

In SPMS analysis, the mtDNA fragment encompassing nt 7362–7580 was PCR-amplified using a 'sense' primer corresponding to the mtDNA sequence from nt 7362 to nt 7371, and an 'antisense' primer, biotinylated at the 5′ end, corresponding to the reverse-complementary mtDNA sequence from nt 7561 to nt 7580. Aliquots (10 µl) of the PCR reaction were incubated for 1.5 h at 37°C in microtitration wells precoated with streptavidin (Labsystems, Finland). After denaturation with 50 mM NaOH, two one-step primer-extention reactions were carried out directly in the micro-wells, using a 'detection' primer corresponding to the 'sense' mtDNA sequence from 7452 to 7471. In one reaction, to detect the wild-type A^{7472}, the radioactive nucleotide species to be incorporated was 0.4 µM [^{3}H]-dATP; in the second reaction, to detect the extra $C^{7472.i}$, the radioactive nucleotide species was 0.4 µM [^{3}H]-dCTP.

After washing with 20 mM Na-phosphate buffer pH 7.5, 0.1% Tween-20, the radiolabeled DNA products of the primer-extension reactions were eluted with 50 mM NaOH. Radioactivity was measured in a liquid scintillation counter and the [^{3}H]-C/[^{3}H]-A ratios were compared with a standard curve obtained by using known proportions of the mutant and wild-type mtDNA fragments as templates for the PCR reactions. To select the mutant and wild-type fragments used to prepare the standard curve, PCR products from the proband's and a control DNA were inserted into the pCR II plasmid of the TA-cloning kit (Invitrogen). Ten clones from the proband's and one clone from the control DNA were then sequenced as described above.

Controls were 381 DNA samples obtained from blood or muscle of as many unrelated individuals. A total of 328 blood DNA samples originated as follows: 300 were from normal individuals of Southern Italy; 14 were from normal individuals of Scotland; three were from Scottish patients affected by sporadic neurosensorial deafness; three were from patients (one Chinese and two Scots) affected by np 1555-positive AAID; 11 were from Chinese patients with np 1555-negative AAID. The 50 muscle DNA samples used as controls were from individuals of different Italian regions, affected by various neuromuscular disorders including mitochondrial encephalomyopathies.

Cell culture studies

Mass myoblast culture was established from the muscle biopsy of patient III-12 (18,30). The King-Attardi protocol was used to obtain transmitochondrial cybrids (15), by PEG-fusing $2–5\times10^5$ enucleated myoblasts with 10^6 143B.TK$^-$-derived rho° cells (18). The polymorphic *Rsa*I restriction site at

mtDNA np 16310 (1,5), contained in the 143B mitochondrial genotype, and absent in the proband's mtDNA, was used to confirm that cybrid mitochondria contained exclusively the latter. Conversely, the hypervariable region in the human apolipoprotein B gene (31) was used to demonstrate that the nuclear genome belonged exclusively to 143B-derived rho° cells.

Computer analysis

Statistical analysis was carried out using the unpaired, two-tailed Student's *t* test and the Mann–Whitney non-parametric test, both contained in the StatView- program for Macintosh (version 4.2).

Prediction of RNA secondary structure and calculation of free energy were carried out using the Zucker algorhythm of the FOLD program, in the GCG-Wisconsin package of softwares. Interspecies alignment of the $tRNA^{Ser(UCN)}$ was obtained using previously published sequences (12).

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